

BluHorizon AI — Full-Stack RAG Application

Overview

BluHorizon AI is a full-stack Retrieval-Augmented Generation (RAG) system with a Flutter mobile frontend and a Python FastAPI backend. Users can ingest documents (PDF, DOCX, XLSX), which are embedded into a FAISS vector store, then query them through an AI-powered chat interface backed by Claude.

Technologies

Layer	Stack
Frontend	Flutter (Dart), mobile-first
Backend	Python, FastAPI, Uvicorn
AI	Anthropic Claude API
Vector DB	FAISS (CPU)
Embeddings	Sentence-Transformers
Database	Firebase, SQLite
ML	PyTorch, tiktoken
Search	Tavily API
Deployment	Docker, Docker Compose

Architecture

```
docker-compose.yml
├─ Backend (port 8000)
│   └─ routers/           ← chat, RAG, agents endpoints
│       └─ services/
│           └─ claude_client.py    ← Anthropic API wrapper (tool
support)
│               └─ rag_pipeline.py    ← HyDE-based RAG pipeline
│                   └─ embedding.py    ← Vector embedding generation
│                       └─ store_faiss_vector.py ← FAISS vector store operations
│                           └─ sqlite.py    ← SQLite integration
├─ Frontend (port 3000)
│   └─ Flutter mobile app
│       └─ ViewModels (GPS, Camera)
```

Key Features

- **HyDE RAG Pipeline:** Generates a hypothetical answer to the user query, embeds it, and uses it for semantic retrieval — improving recall over naïve keyword search
- **Gap-aware chunk filtering:** Dynamic score-gap cutoff selects the most relevant document chunks rather than using a fixed top-K
- **Document ingestion:** Supports PDF, DOCX, and XLSX with configurable chunking strategies
- **Tool-augmented Claude:** Custom tools (web search, code execution) extend Claude's capabilities within the chat interface
- **Mobile-native features:** GPS and camera ViewModels integrated into the Flutter frontend
- **Containerized deployment:** Single docker-compose up launches both backend and frontend

Team Contributions

- **Diego:** ViewModels (GPS and Camera integration)
- **Payton:** UI/UX design
- **Jacky + Connor:** Firebase / database layer

RAG Pipeline Flow

User query

- HyDE: Claude generates hypothetical document
- Embed hypothetical doc with Sentence-Transformers
- FAISS similarity search
- Gap-aware chunk selection
- Retrieved chunks + original query → Claude response